MIL-D-18506(NOrd)
25 April 1955
SUPERSEDING
NAVORD OS 1599
18 January 1952

MILITARY SPECIFICATION

DETONATOR MK 45 MOD 0

1. SCOPE

1.1 This specification governs the manufacture, assembly, loading, and preparation for delivery of Detonator Mk 45 Mod 0 and the methods of inspection and tests upon which acceptance will be based. The Detonator Mk 45 Mod 0 is flash initiated.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

JAN-T-339 -- Tetryl (trinitrophenylmethylnitramine).

STANDARDS

MIL-STD-105-- Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129-- Marking for Shipment and Storage

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DRAWINGS

BUREAU OF ORDNANCE

LD 166550 -- Primer (Fuze, Electric) Mk 112 Mod 0,
LD 166576 -- Detonator Mk 45 Mod 0,
LD 168793 -- Detonator Container Mk 3 Mod 0,
LD 168794 -- Detonator Container Mk 2 Mod 0,
LD 185941 -- Testing Apparatus, and all documents listed thereon.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 -- General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. -- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 -- Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950).")

3. REQUIREMENTS

3.1 General.

3.1.1 Detonators Mk 45 Mod 0 covered by this specification shall be manufactured in accordance with Bureau

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of Ordnance List of Drawings LD No. 166576 and all documents listed thereon.

3.1.2 The requirements and procedures established in OP 400 shall be adhered to wherever applicable.

3.2 Lead Azide. -- The moisture content of the lead azide at time of loading shall be less than 0.30 percent when tested as specified in 4.4.

3.3 Tetryl. -- The volatiles and moisture content of the tetryl at time of loading shall be less than 0.10 percent when tested as specified in 4.5.

3.4 Functioning. -- Each lot of detonators shall pass the functioning test specified in 4.6.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Preproduction Sample. -- Before entering into quantity production a preproduction sample of two hundred (200) Detonators Mk 45 Mod 0 shall be delivered by the Contractor for test to an agency designated by the Procuring Agency. These detonators shall be manufactured by the same procedures and processes and at the same location proposed by the Contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended the Contractor will be notified and will be authorized by the Procuring Agency to proceed with the submission of the product for acceptance. Accepted samples will become the property of the Procuring Agency and will be included in the number called for in the contract schedule.

4.1.1 Preproduction Sample for a Subsequent Contract. -- The necessity for a preproduction sample will be determined by the Procuring Agency when production under a new contract by the same Contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

4.2 Production Lot. -- A production lot shall consist of 10,000 detonators. The detonators of a lot shall be of consecutive manufacture. The production lot is used for delivery and identification of detonators and is not related to the inspection lot defined in Military Standard MIL-STD-105.

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4.3 Inspection. -- The general inspection procedures shall be in accordance with General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment, OP 400; government inspection shall be in accordance with the provisions of General Procedure for Government Inspection and contractors inspection shall be in accordance with Inspection by Contractor, as contained in OP 400.

4.4 Moisture Content Test for Lead Azide. -- Conduct at least one test for each lot of detonators and such additional tests as the Inspector considers warranted by variations in the processes and materials. Collect an approximate two gram sample in a tared, stoppered container at the site of and immediately prior to consolidating the charge in the detonator. Weigh the container; open and expose the sample for drying at 50 ± 2 degrees C and at least 28 inches (Hg) vacuum until constant weight is reached. Cool the container in a desiccator; stopper and weigh. All weighings shall be accurate to ± 0.5 milligram. The presence of moisture in excess of 0.30 percent shall cause suspension of the charge loading until permanent corrective measures satisfactory to the Inspector have been completed. Detonators which have been loaded with defective charges shall be rejected .

4.5 Volatiles and Moisture Content Test for Tetryl. -- Conduct at least one test for each lot of detonators and such additional tests as the Inspector considers warranted by variations in the process and materials. Test procedure shall be in accordance with that given in Military Specification JAN-T-339. The presence of volatiles and moisture in excess of 0.10 percent shall cause suspension of the charge loading until permanent corrective measures satisfactory to the Inspector have been completed. Detonators which have been loaded with defective charges shall be rejected.

4.6 Functioning Test. -- Select at random from each lot, 85 detonators for the functioning test specified in 4.6.1. The average block expansion ($\bar{x}$) and the standard deviation (s) of the expansion values in inches shall be calculated in accordance with the provisions of NAVORD Form 1844C (New 6/51). The lot shall be rejected if $\bar{x}$ minus 2s is less less than 0.085 inch.

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4.6.1 Test Procedure. -- Test the sample detonators in the Test Apparatus for the Mk 45 Detonators (LD 185941). The test apparatus shall be situated in an armored cubicle equipped with a six volt electrical line controlled by an external firing switch. Before firing each detonator measure the diameter of the Holder (SK 123185) in at least two places and record the largest value. Insert each detonator in a measured holder and mount it in the test apparatus described on LD 185941. Fire the detonator using a Primer Mk 112 Mod 0 (LD 166550). After firing repeat the diameter measurements of the holder. Again measure in at least two places and record the largest measurement. The difference in the holder diameter as measured before and after firing is the expansion of the holder and is the value "$\bar{x}$" in inches used on NAVORD Form 1844C (New 6/51). In this test no holder shall be reused.

5. PREPARATION FOR DELIVERY

5.1 Packaging. -- One hundred (100) Detonators Mk 45 Mod 0 shall be packaged in the metal container specified on Bureau of Ordnance List of Drawings LD No. 168793.

5.2 Packing. -- Five thousand (5000) Detonators Mk 45 Mod 0, packaged as specified in 5.1, shall be packed in accordance with the drawings listed on Bureau of Ordnance List of Drawings LD No. 168794.

5.3 Marking. -- In addition to any special marking required by the contract or order, all containers shall be marked in accordance with Military Standard MIL-STD-129 and Code of Federal Regulations 49 CFR 71-78.

5.4 Data Cards. -- Six data cards, five by eight inches, printed as shown below shall be filled out for each lot of detonators. Two of these cards shall accompany the shipping ticket, three shall be forwarded to the Bureau of Ordnance, and one shall be furnished to the Inspector.

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AMMUNITION DATA CARD

Mark No. _____ Lot No. _____

Dwg. No. _____ Spec. No. _____

Loaded By _____ Date (month and year)____

Contract No. _____ Req. No. _____

No. in box _____ No. in lot _____

Computations of Data (Test Apparatus - LD No. 185491)

Mean $\bar{x}$ = _____ Std. Deviation s = _____

:MATERIAL	:MANUFACTURER:	LOT NO.:	DATE (MONTH AND YEAR):
SEALING DISC			
CUP			
LEAD AZIDE			
TETRYL			

Remarks: (These are to include any changes in the manufacturing processes which the Inspector feels might destroy the uniformity of the lot.)

DATA CARD

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6. NOTES. -- There are none applicable.

Patent Notice. -- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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